

# PE-EC603B — Bio-Medical Electronics

Semester VI • 3L:0T:0P • 3 Credits

## Module 1 Exam Notes: Introduction to Human Physiology

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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### Contents

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|          |                                                               |          |
|----------|---------------------------------------------------------------|----------|
| <b>1</b> | <b>The Human Body as an Engineering System</b>                | <b>2</b> |
| 1.1      | Organ Systems Overview                                        | 2        |
| 1.2      | Homeostasis and Feedback Control                              | 2        |
| <b>2</b> | <b>The Cardiovascular System</b>                              | <b>3</b> |
| 2.1      | Structure and Function                                        | 3        |
| 2.2      | The Cardiac Cycle and Electrical Conduction                   | 3        |
| 2.3      | The ECG Signal                                                | 4        |
| <b>3</b> | <b>The Respiratory System</b>                                 | <b>4</b> |
| 3.1      | Structure and Function                                        | 5        |
| 3.2      | Lung Volumes and Capacities                                   | 5        |
| 3.3      | Respiratory Rate and Clinical Monitoring                      | 5        |
| <b>4</b> | <b>The Nervous System</b>                                     | <b>6</b> |
| 4.1      | Organisation                                                  | 6        |
| 4.2      | The Neuron and Action Potential                               | 6        |
| 4.3      | The Brain and EEG Signals                                     | 7        |
| <b>5</b> | <b>The Muscular System and Bio-Electrical Signals Summary</b> | <b>7</b> |
| 5.1      | Summary of Bio-Electrical Signals                             | 8        |

## 1. The Human Body as an Engineering System

The human body can be viewed as a highly complex, self-regulating biological machine. From a bio-medical electronics perspective, it is a system with sensors (sense organs), processors (nervous system and brain), actuators (muscles), and a power supply (metabolic energy from food). Understanding this system is essential for designing instruments that measure, monitor, and assist bodily functions.

### 1.1 Organ Systems Overview

#### Definition: Organ System

An **organ system** is a group of organs that work together to perform one or more physiological functions. The human body consists of 11 major organ systems, each with specific roles in maintaining homeostasis.

#### Major Organ Systems Relevant to Bio-Medical Electronics:

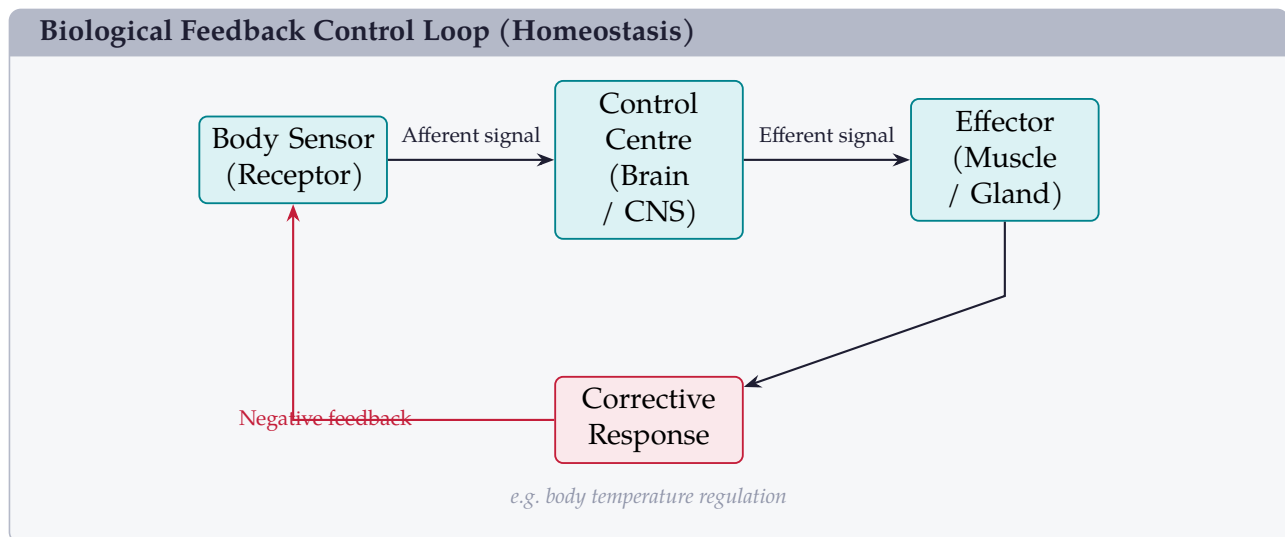
| System          | Primary Organs                         | Clinical Relevance                                 |
|-----------------|----------------------------------------|----------------------------------------------------|
| Cardiovascular  | Heart, arteries, veins, capillaries.   | ECG, blood pressure, cardiac output monitoring.    |
| Respiratory     | Lungs, diaphragm, trachea, bronchi.    | Spirometry, pulse oximetry, ventilator design.     |
| Nervous         | Brain, spinal cord, peripheral nerves. | EEG, EMG, nerve conduction velocity.               |
| Muscular        | Skeletal, smooth, cardiac muscle.      | EMG, prosthetic limb control, physiotherapy.       |
| Renal (Urinary) | Kidneys, ureter, bladder.              | Dialysis machine design, fluid balance monitoring. |
| Endocrine       | Pancreas, thyroid, adrenal glands.     | Blood glucose monitoring, hormone assay.           |

### 1.2 Homeostasis and Feedback Control

#### Definition: Homeostasis

**Homeostasis** is the ability of a living organism to maintain a stable internal environment despite changes in external conditions. It is achieved through continuous negative feedback loops involving sensors, control centers, and effectors.

The body maintains homeostasis by regulating parameters such as body temperature ( $37 \pm 0.5^\circ\text{C}$ ), blood pH (7.35–7.45), blood glucose (70–100 mg/dL), and arterial blood pressure (120/80 mmHg). Most clinical instruments are designed to monitor deviations from these normal (baseline) values.



## 2. The Cardiovascular System

The cardiovascular system (CVS) is the most extensively monitored system in bio-medical electronics, since the heart generates measurable electrical, pressure, and acoustic signals accessible from the body surface.

### 2.1 Structure and Function

#### Definition: Cardiovascular System

The **cardiovascular system** consists of the heart (pump), blood vessels (conduits), and blood (transport medium). It delivers oxygen, nutrients, and hormones to all tissues and removes metabolic waste products.

The heart is a four-chambered muscular pump:

- **Right Atrium (RA):** Receives deoxygenated blood from the body via the superior and inferior vena cava.
- **Right Ventricle (RV):** Pumps blood to the lungs through the pulmonary artery (pulmonary circulation).
- **Left Atrium (LA):** Receives oxygenated blood from the lungs via pulmonary veins.
- **Left Ventricle (LV):** Pumps oxygenated blood to the entire body through the aorta (systemic circulation). The LV wall is thickest because it works against the highest pressure.

### 2.2 The Cardiac Cycle and Electrical Conduction

Each heartbeat is initiated by the **Sinoatrial (SA) node** — the heart's natural pacemaker located in the right atrium. The electrical impulse propagates through a defined conduction pathway:

**SA Node → AV Node → Bundle of His → Left and Right Bundle Branches → Purkinje Fibres → Ventricular Myocardium**

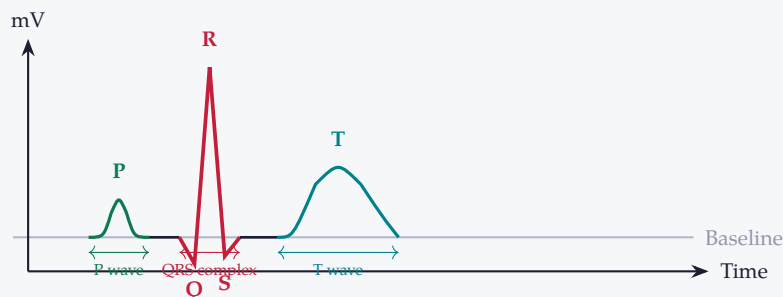
### Cardiovascular Parameters and Normal Values

|                               |                                                                     |
|-------------------------------|---------------------------------------------------------------------|
| <b>Heart Rate (HR)</b>        | 60–100 beats/min (normal); < 60 = bradycardia; > 100 = tachycardia. |
| <b>Stroke Volume (SV)</b>     | Volume ejected per beat $\approx$ 70 mL at rest.                    |
| <b>Cardiac Output (CO)</b>    | $CO = HR \times SV \approx$ 5 L/min at rest.                        |
| <b>Blood Pressure (BP)</b>    | Systolic $\approx$ 120 mmHg; Diastolic $\approx$ 80 mmHg.           |
| <b>Mean Arterial Pressure</b> | $MAP = \frac{SBP + 2 \times DBP}{3} \approx$ 93 mmHg.               |

## 2.3 The ECG Signal

The electrocardiogram (ECG) is a recording of the electrical activity of the heart from the body surface. Each cardiac cycle produces a characteristic waveform:

### ECG Waveform — Components and Significance



### ECG Wave Significance:

| Wave               | Physiological Event                                             | Normal Duration / Amplitude |
|--------------------|-----------------------------------------------------------------|-----------------------------|
| <b>P wave</b>      | Atrial depolarization (SA node fires, atria contract).          | 80–100 ms; < 0.25 mV.       |
| <b>PR Interval</b> | Time from atrial to ventricular depolarization (AV node delay). | 120–200 ms.                 |
| <b>QRS Complex</b> | Ventricular depolarization (ventricles contract).               | < 120 ms; R up to 1.5 mV.   |
| <b>ST Segment</b>  | Ventricular plateau (all cells depolarized, isoelectric).       | Flat at baseline.           |
| <b>T wave</b>      | Ventricular repolarization (ventricles relax).                  | 160 ms; 0.1–0.5 mV.         |

## 3. The Respiratory System

### 3.1 Structure and Function

#### Definition: Respiratory System

The **respiratory system** is responsible for gas exchange between the body and the environment — bringing oxygen ( $O_2$ ) into the blood and expelling carbon dioxide ( $CO_2$ ). It consists of the conducting airways (nose, pharynx, trachea, bronchi) and the respiratory zone (alveoli where gas exchange occurs).

#### Mechanics of Breathing:

- **Inspiration (active):** Diaphragm and external intercostals contract → thoracic cavity volume increases → intrapulmonary pressure drops below atmospheric pressure → air flows into lungs.
- **Expiration (passive at rest):** Diaphragm relaxes → thoracic cavity recoils → pressure rises above atmospheric → air flows out.

### 3.2 Lung Volumes and Capacities

#### Lung Volume Parameters

|                                         |                                                                               |
|-----------------------------------------|-------------------------------------------------------------------------------|
| <b>Tidal Volume (TV)</b>                | Volume inhaled/exhaled in one normal breath $\approx 500$ mL.                 |
| <b>Inspiratory Reserve Volume (IRV)</b> | Extra air that can be inhaled above TV $\approx 3000$ mL.                     |
| <b>Expiratory Reserve Volume (ERV)</b>  | Extra air exhaled forcefully below TV $\approx 1100$ mL.                      |
| <b>Residual Volume (RV)</b>             | Air remaining after maximal expiration $\approx 1200$ mL (cannot be exhaled). |
| <b>Total Lung Capacity (TLC)</b>        | $TLC = TV + IRV + ERV + RV \approx 5800$ mL.                                  |
| <b>Vital Capacity (VC)</b>              | $VC = TV + IRV + ERV \approx 4600$ mL (measurable).                           |
| <b>FEV<sub>1</sub></b>                  | Forced Expiratory Volume in 1 second; $FEV_1/VC > 70\%$ is normal.            |

### 3.3 Respiratory Rate and Clinical Monitoring

Normal respiratory rate (RR) is 12–20 breaths/min in adults. Clinically monitored parameters include:

- **Pulse Oximetry ( $SpO_2$ ):** Non-invasive measurement of blood oxygen saturation using red and infrared LEDs. Normal  $SpO_2 \geq 95\%$ .
- **Capnography ( $EtCO_2$ ):** Monitors exhaled  $CO_2$  concentration,  $\approx 35$ – $45$  mmHg at end-expiration.
- **Spirometry:** Measures lung volumes and flow rates to diagnose obstructive (asthma, COPD) and restrictive disorders.

## 4. The Nervous System

### 4.1 Organisation

#### Definition: Nervous System

The **nervous system** is the body's primary communication and control network. It collects sensory information, processes it, and coordinates responses. It is divided into the **Central Nervous System (CNS)** — brain and spinal cord — and the **Peripheral Nervous System (PNS)** — all nerves outside the CNS.

### 4.2 The Neuron and Action Potential

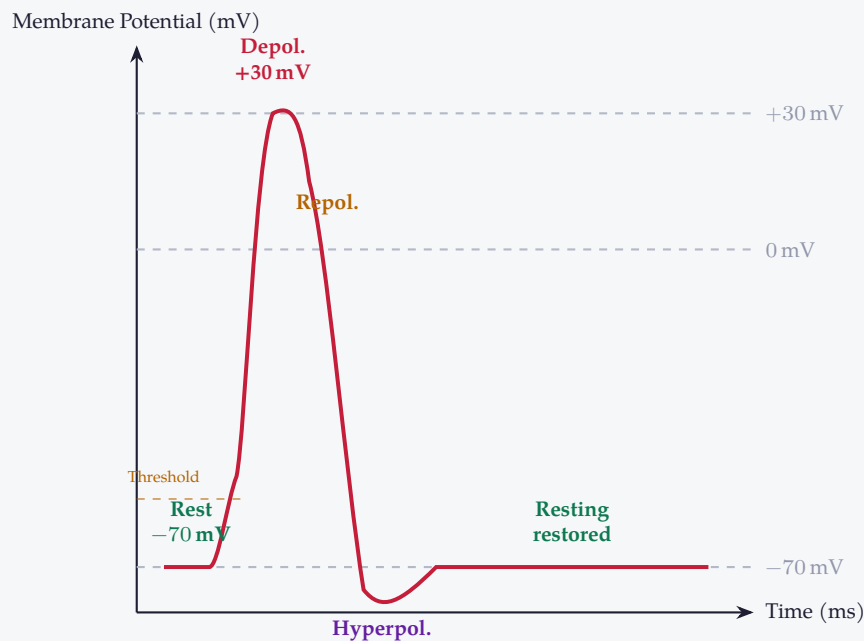
The basic functional unit of the nervous system is the **neuron (nerve cell)**. It consists of:

- **Dendrites:** Short branching processes that receive incoming signals from other neurons.
- **Cell Body (Soma):** Contains the nucleus and integrates incoming signals.
- **Axon:** Long fibre that carries signals (action potentials) away from the cell body to the next neuron or effector.
- **Myelin Sheath:** Insulating layer (formed by Schwann cells in PNS) that speeds up signal conduction via saltatory propagation. Conduction velocity in myelinated fibres: 40–70 m/s.

#### The Action Potential — Phases and Ion Movements

|                           |                                                                                                                 |
|---------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Resting Potential</b>  | Membrane potential $\approx -70$ mV. Inside negative due to $K^+$ leak and $Na^+/K^+$ pump.                     |
| <b>Depolarization</b>     | $Na^+$ channels open rapidly $\rightarrow Na^+$ rushes in $\rightarrow$ membrane potential rises to $+30$ mV.   |
| <b>Repolarization</b>     | $Na^+$ channels inactivate; $K^+$ channels open $\rightarrow K^+$ flows out $\rightarrow$ membrane repolarizes. |
| <b>Hyperpolarization</b>  | $K^+$ channels close slowly $\rightarrow$ brief dip below $-70$ mV (refractory period).                         |
| <b>All-or-Nothing Law</b> | An action potential either fires fully or not at all; amplitude does not vary with stimulus strength.           |

## Action Potential Waveform



## 4.3 The Brain and EEG Signals

The brain generates rhythmic electrical activity that can be recorded as an **Electroencephalogram (EEG)** from scalp electrodes. Different frequency bands correspond to different states of consciousness:

| Wave               | Freq. (Hz) | Amplitude ( $\mu\text{V}$ ) | State                                  |
|--------------------|------------|-----------------------------|----------------------------------------|
| Delta ( $\delta$ ) | 0.5 – 4    | 20 – 200                    | Deep sleep (slow-wave sleep).          |
| Theta ( $\theta$ ) | 4 – 8      | 20 – 100                    | Drowsiness, light sleep, creativity.   |
| Alpha ( $\alpha$ ) | 8 – 13     | 20 – 60                     | Relaxed wakefulness, eyes closed.      |
| Beta ( $\beta$ )   | 13 – 30    | 5 – 30                      | Active thinking, concentration, alert. |
| Gamma ( $\gamma$ ) | 30 – 100   | < 10                        | Intense cognitive processing.          |

## 5. The Muscular System and Bio-Electrical Signals Summary

## Types of Muscle Tissue

- **Skeletal Muscle:** Voluntary, striated, attached to bones. Controlled by the somatic nervous system. Generates **EMG** signals.
- **Cardiac Muscle:** Involuntary, striated, forms the heart wall. Self-excitable via SA node. Generates **ECG** signals.
- **Smooth Muscle:** Involuntary, non-striated, found in blood vessel walls, digestive tract, lungs. Controlled by the autonomic nervous system.

## 5.1 Summary of Bio-Electrical Signals

| Signal | Physiological Origin                                   | Amplitude       | Frequency Range |
|--------|--------------------------------------------------------|-----------------|-----------------|
| ECG    | Cardiac muscle depolarization/repolarization.          | 0.5 – 4 mV      | 0.05 – 150 Hz   |
| EMG    | Skeletal muscle fibre action potentials (motor units). | 0.1 – 5 mV      | 5 – 2000 Hz     |
| EEG    | Aggregate cortical neuron activity (brain).            | 5 – 300 $\mu$ V | 0.5 – 100 Hz    |
| EOG    | Retinal dipole movement with eye motion.               | 0.5 – 1 mV      | DC – 100 Hz     |
| ERG    | Retinal response to light stimulation.                 | 0.5 – 1 mV      | DC – 50 Hz      |
| EKG    | Gastro-intestinal smooth muscle electrical activity.   | 0.1 – 1 mV      | 0.05 – 1 Hz     |



## Quick Revision — Module 1

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### Key Formulas and Normal Values

|                                        |                                                                                    |
|----------------------------------------|------------------------------------------------------------------------------------|
| <b>Cardiac Output</b>                  | $CO = HR \times SV \quad (\approx 5 \text{ L/min at rest})$                        |
| <b>Mean Arterial Pressure</b>          | $MAP = \frac{SBP + 2 \times DBP}{3}$                                               |
| <b>Total Lung Capacity</b>             | $TLC = TV + IRV + ERV + RV \approx 5800 \text{ mL}$                                |
| <b>Vital Capacity</b>                  | $VC = TV + IRV + ERV \approx 4600 \text{ mL}$                                      |
| <b>Nernst Equation (ion potential)</b> | $E = \frac{RT}{zF} \ln \frac{[\text{ion}]_{\text{out}}}{[\text{ion}]_{\text{in}}}$ |
| <b>Resting Membrane Potential</b>      | $\approx -70 \text{ mV (inside negative)}$                                         |

### One-Line Definitions

- **Homeostasis:** The body's ability to maintain a stable internal environment via negative feedback loops.
- **SA Node:** Sinoatrial node; the heart's natural pacemaker located in the right atrium; fires at 60–100 bpm.
- **Action Potential:** A brief, self-propagating reversal of membrane polarity ( $-70$  to  $+30 \text{ mV}$ ) that transmits nerve signals.
- **Cardiac Output:** Volume of blood pumped per minute by either ventricle;  $CO = HR \times SV$ .
- **Tidal Volume:** Volume of air inhaled or exhaled in one normal, quiet breath ( $\approx 500 \text{ mL}$ ).
- **Vital Capacity:** Maximum volume of air a person can expire after a maximal inspiration.
- **Depolarization:** Rapid influx of  $\text{Na}^+$  ions causing the cell membrane to become positive inside.
- **ECG:** Electrocardiogram; surface recording of the heart's electrical activity.
- **EEG:** Electroencephalogram; scalp recording of brain electrical activity.
- **EMG:** Electromyogram; recording of skeletal muscle electrical activity.

### Frequently Asked / Viva Questions

**Q1. What is homeostasis? Give an example.**

Homeostasis is the maintenance of a stable internal environment through

**Frequently Asked / Viva Questions (contd.)**

negative feedback. Example: If body temperature rises, sweat glands are activated to cool the body back to 37 °C.

**Q2. Describe the role of the SA node in the heart.**

The SA node is the natural cardiac pacemaker located in the right atrium. It spontaneously depolarizes at 60–100 bpm, initiating an electrical impulse that propagates to the AV node, bundle of His, and Purkinje fibres, triggering coordinated ventricular contraction.

**Q3. What physiological event does the QRS complex of the ECG represent?**

The QRS complex represents ventricular depolarization — the electrical event that triggers the simultaneous contraction of the left and right ventricles.

**Q4. Define cardiac output and state its normal value.**

Cardiac Output (CO) is the volume of blood ejected by one ventricle per minute:  $CO = HR \times SV \approx 5 \text{ L/min}$  at rest for an adult.

**Q5. What is the difference between systolic and diastolic blood pressure?**

Systolic pressure ( $\approx 120 \text{ mmHg}$ ) is the peak pressure in the arteries during ventricular contraction. Diastolic pressure ( $\approx 80 \text{ mmHg}$ ) is the minimum pressure during ventricular relaxation.

**Q6. Explain the all-or-nothing law of nerve conduction.**

Once a stimulus exceeds the threshold (about  $-55 \text{ mV}$ ), a full action potential of fixed amplitude is triggered. A sub-threshold stimulus produces no action potential at all. The amplitude does not increase with stronger stimuli.

**Q7. List the EEG wave bands and their corresponding brain states.**

Delta (0.5–4 Hz): deep sleep; Theta (4–8 Hz): drowsiness; Alpha (8–13 Hz): relaxed wakefulness; Beta (13–30 Hz): active alertness; Gamma ( $> 30 \text{ Hz}$ ): intense cognition.

**Q8. What is the difference between ECG, EMG, and EEG signals in terms of amplitude and frequency?**

ECG: 0.5–4 mV, 0.05–150 Hz (cardiac). EMG: 0.1–5 mV, 5–2000 Hz (skeletal muscle). EEG: 5–300  $\mu\text{V}$ , 0.5–100 Hz (brain) — much smaller amplitude than ECG or EMG.

**Q9. What is the role of the myelin sheath in nerve conduction?**

The myelin sheath acts as an electrical insulator around the axon. Depolarization jumps between nodes of Ranvier (saltatory conduction), dramatically increasing conduction velocity to 40–70 m/s compared to 0.5–2 m/s in unmyelinated fibres.

**Q10. Define tidal volume and vital capacity.**

Tidal Volume (TV) is the volume of air inhaled/exhaled per normal breath ( $\approx 500 \text{ mL}$ ). Vital Capacity (VC) is the maximum air that can be expelled after maximum inhalation ( $VC = TV + IRV + ERV \approx 4600 \text{ mL}$ ).

## Frequently Asked / Viva Questions (contd.)

**Q11. Why is the left ventricular wall thicker than the right?**

The left ventricle pumps blood through the systemic circulation (entire body), working against a much higher resistance (pressure  $\approx 120$  mmHg). The right ventricle only pumps to the lungs (pulmonary pressure  $\approx 25$  mmHg), so it needs less force.

## Bio-Electrical Signal Comparison

| Signal       | ECG                  | EMG             | EEG                       |
|--------------|----------------------|-----------------|---------------------------|
| Source       | Heart muscle         | Skeletal muscle | Brain cortex              |
| Amplitude    | 0.5 – 4 mV           | 0.1 – 5 mV      | 5 – 300 $\mu$ V           |
| Bandwidth    | 0.05 – 150 Hz        | 5 – 2000 Hz     | 0.5 – 100 Hz              |
| Electrodes   | Surface, chest       | Surface/needle  | Scalp (10-20 system)      |
| Clinical use | Arrhythmia, ischemia | Neuropathy, NCV | Epilepsy, sleep disorders |